

Yersinia enterocolitica Infectious Disease (Yersiniosis): A Comprehensive Disease Characterization

Disease: Yersinia enterocolitica infectious disease (yersiniosis) **MONDO ID:** MONDO:0042370 · **ICD-10:** A04.6 · **ICD-11:** 1A03 · **MeSH:** D015008 (Yersinia enterocolitica infection) / D015009 (Yersiniosis) · **Pathogen NCBI Taxon:** NCBI:txid630 · **Category:** Infectious Disease

Summary

Yersinia enterocolitica infectious disease (yersiniosis) is a **foodborne zoonotic enteric infection** caused by the Gram-negative coccobacillus *Yersinia enterocolitica*. It is most commonly acquired through consumption of undercooked or contaminated **pork**, with swine serving as the principal reservoir of the human-pathogenic bioserotype 4/O:3 ([PMID: 26403101](#), [PMID: 29307129](#)). The organism is **psychrotolerant** (grows at 4 °C) and forms biofilms throughout the cold food chain, making it a persistent hazard in refrigerated foods, dairy, and blood products ([PMID: 41539755](#)). Non-pestis yersiniosis has shown the largest increase in reported incidence of any bacterial enteric infection in the United States during 2016–2023 ([PMID: 42720049](#)).

Mechanistically, after ingestion the bacterium colonizes the terminal ileum, uses the outer-membrane adhesin **invasin** to bind host **β1 integrins** on **M cells** overlying Peyer's patches, and invades follicle-associated lymphoid tissue ([PMID: 27107739](#)). At 37 °C the ~70-kb **pYV virulence plasmid** activates (via the VirF/LcrF regulator) a **type III secretion system (T3SS)** that injects **Yop effector proteins** into phagocytes, subverting phagocytosis, actin dynamics, and NF-κB/MAPK survival and inflammasome signaling — YopP/YopJ in particular drives macrophage/dendritic-cell death ([PMID: 22563435](#)). Host **iron status** is a decisive branch point: because *Y. enterocolitica* lacks its own high-affinity siderophore, iron overload and **deferoxamine** therapy dramatically increase virulence and the risk of septicemia, lowering the murine LD₅₀ by more than 5 log units ([PMID: 12019084](#)).

The clinical spectrum is **age- and host-dependent**: self-limiting enterocolitis in young children; pseudoappendicitis (terminal ileitis / mesenteric adenitis) in older children and young adults; and invasive septicemia with hepatic/splenic abscesses in iron-overloaded or immunocompromised hosts (PMID: 15254824, PMID: 9376035). Important **post-infectious sequelae** include HLA-B27-associated **reactive arthritis**, erythema nodosum, uveitis (PMID: 40473988, PMID: 7221767), and **autoimmune thyroid disease / Graves' disease** via molecular mimicry between the bacterial outer-membrane porin OmpF and the TSH receptor (PMID: 32564766, PMID: 20484489). Most cases resolve without antibiotics; invasive disease is treated with third-generation cephalosporins, fluoroquinolones, and/or aminoglycosides — the organism is intrinsically resistant to ampicillin and first-generation cephalosporins (PMID: 10803262). No human vaccine exists; prevention relies on farm-to-fork food-chain hygiene (PMID: 19058737).

1. Disease Information

Overview. Yersiniosis is an enteric infection caused by *Yersinia enterocolitica*, a Gram-negative, facultatively anaerobic, non-lactose-fermenting member of the Enterobacteriaceae. It is one of the more common causes of bacterial gastroenteritis in temperate regions and is notable for its psychrotolerance and its zoonotic, pork-associated transmission. Clinically it manifests along a spectrum from acute enteritis/enterocolitis to pseudoappendicitis and, in vulnerable hosts, septicemia (PMID: 15254824).

Key identifiers. MONDO:0042370; ICD-10 A04.6 (Enteritis due to *Yersinia enterocolitica*); ICD-11 1A03 (Enteritis due to *Yersinia enterocolitica*); MeSH D015008/D015009; causative organism NCBI:txid630.

Synonyms / alternative names. Yersiniosis; intestinal yersiniosis; *Yersinia enterocolitica* enteritis/enterocolitis; enteric yersiniosis; non-pestis yersiniosis.

Data provenance. This report aggregates disease-level evidence from primary literature, microbiological surveillance, case reports, meta-analysis, and experimental (rodent/in-vitro) studies rather than individual EHR records.

2. Etiology

Causal factor (infectious). The disease is caused by pathogenic bioserotypes of *Y. enterocolitica*. Globally, the principal human-pathogenic serotypes are **O:3, O:9, O:8, and O:5,27**, with bioserotype **4/O:3** dominant in most regions ([PMID: 29307129](#), [PMID: 40445798](#)). Pigs are the primary reservoir; insufficiently cooked pork is the main vehicle ([PMID: 26403101](#)).

Environmental / exposure risk factors. Consumption of raw or undercooked pork (including chitterlings), unpasteurized dairy, and contaminated water; cold-chain storage that permits psychrotolerant growth; transfusion via contaminated blood products; occupational contact in the pork production chain. Young age is a risk factor for enterocolitis.

Host risk factors (the key modifier — iron). Iron overload and iron-chelation therapy with **deferoxamine** are the most important host risk factors for severe/invasive disease. In a mouse model, deferoxamine pretreatment decreased the LD₅₀ of *Y. enterocolitica* by **>5 log units**, whereas deferiprone did not ([PMID: 12019084](#)). Transfusion-dependent conditions (β -thalassemia, sickle cell disease) and dialysis iron supplementation predispose to fulminant infection ([PMID: 31219970](#), [PMID: 21266629](#), [PMID: 11698009](#), [PMID: 21151549](#)). Diabetes, malignancy, and other immunocompromising conditions also predispose to bacteremia ([PMID: 9376035](#)).

Genetic host factor. **HLA-B27** is a genetic susceptibility factor for the post-infectious complication of reactive arthritis ([PMID: 7221767](#)).

Protective factors. Thorough cooking of pork, pasteurization, cold-chain hygiene, and biosecurity reduce exposure. Mucosal innate defenses (e.g., Paneth-cell α -defensin HD6, which entraps enteric pathogens in self-assembled nanonets) limit invasion ([PMID: 22722251](#)). Use of the oral iron chelators **deferiprone/deferasirox** instead of deferoxamine avoids promoting bacterial growth ([PMID: 12019084](#), [PMID: 19413741](#)).

Gene–environment interaction. The clearest interaction is **iron availability × siderophore biology**: exogenous ferrioxamine (from deferoxamine) supplies iron the bacterium cannot otherwise scavenge, converting a contained mucosal infection into systemic disease. A second interaction is **HLA-B27 × enteric infection**, where a genetically predisposed host mounts an aberrant post-infectious arthritic response.

3. Phenotypes

Phenotype	Type	Onset / course	Frequency	HPO suggestion
Diarrhea (often watery, sometimes bloody)	Symptom	Acute, self-limiting	Very common in enterocolitis	HP:0002014
Abdominal pain (right-lower-quadrant)	Symptom	Acute	Common; prominent in pseudoappendicitis	HP:0002027 / HP:0002574
Fever	Sign	Acute	Common	HP:0001945
Terminal ileitis / mesenteric lymphadenitis	Clinical sign	Subacute	Older children/adults	HP:0002583 (enterocolitis)
Pseudoappendicitis	Clinical syndrome	Acute	Older children/young adults	HP:0002605 (approximate)
Septicemia / bacteremia	Lab/clinical	Acute, severe	Rare; iron-overloaded/immunocompromised	HP:0100806
Hepatic/splenic abscesses	Manifestation	Invasive disease	Rare	HP:0012199 / HP:0020104
Reactive arthritis (asymmetric, lower-limb)	Post-infectious	Days–weeks after infection	~1/1000 (ReA overall); young adults 18–40	HP:0001369
Erythema nodosum	Post-infectious skin sign	Weeks after	Occasional	HP:0012219
Uveitis	Post-infectious	Weeks after	Occasional	HP:0000554
Graves' disease / autoimmune thyroiditis	Post-infectious autoimmune	Delayed	Association (see §5)	HP:0100647

Characteristics. Onset is typically **acute**; enterocolitis is generally **self-limiting** (1–3 weeks). Severity ranges from mild (children) to life-threatening (septicemia in iron overload). Post-infectious immunological syndromes are episodic/delayed ([PMID: 8363822](#), [PMID: 3545747](#)).

Quality-of-life impact. Acute enterocolitis causes short-term disability from diarrhea/pain; pseudoappendicitis may lead to unnecessary appendectomy. Reactive arthritis and other post-infectious syndromes can cause prolonged joint pain and functional impairment persisting weeks to months ([PMID: 32648451](#)).

4. Genetic / Molecular Information

Yersiniosis is an infectious, not a Mendelian, disease; there are no causal human genes. The relevant genetics are (a) **host susceptibility loci** and (b) **bacterial virulence genes**.

Host genetics. HLA-B27 predisposes to post-infectious reactive arthritis ([PMID: 7221767](#)). No causal germline variants, chromosomal abnormalities, or pathogenic ACMG-classified variants apply to the disease itself. gnomAD/ClinVar entries are **not applicable**.

Bacterial virulence genetics. Pathogenicity of biotypes 1B and 2–5 depends on **both chromosomal and pYV plasmid-borne genes**; biotype 1A lacks pYV ([PMID: 28400007](#)).

- **Chromosomal:** *ail* (attachment-invasion locus; principal chromosomal virulence marker — [PMID: 28283072](#)); *inv* (invasin); *ystA* (heat-stable enterotoxin YstA; best marker of pathogenic biotypes — [PMID: 28400007](#)); *myfA* (fimbriae); *hreP* (protease). Biotype 1A carries *ystB* and *hreP* but lacks *ail/ystA* ([PMID: 39805396](#)).
- **Plasmid (pYV, ~70 kb):** *virF/lcrF* (master transcriptional activator of the T3SS regulon), *yadA* (adhesin), and the *ysc* operon encoding the T3SS (e.g., *yscE*, *F*, *G*, *I*, *J*, *K*, *H*) plus *yop* effector genes ([PMID: 8709853](#)). YstA enterotoxin production is **pYV-independent** ([PMID: 30961808](#)).

5. Environmental Information

Infectious agent. *Yersinia enterocolitica* (NCBI:txid630); a species divided into ~6 biotypes and ~60 O-serotypes, with O:3, O:9, O:8, O:5,27 principally associated with human disease ([PMID: 40445798](#)).

Environmental factors. The organism's **psychrotolerance** (growth at 4 °C) and **biofilm formation** enable persistence across refrigerated foods and food-processing environments; slaughterhouses (40.2%) and farmers' markets (34.8%) are high-risk contamination zones (PMID: 41539755). Contaminated red-blood-cell units support proliferation during cold storage, a recognized transfusion hazard (PMID: 33341965).

Lifestyle / dietary factors. Consumption of raw/undercooked pork (notably chitterlings), unpasteurized milk/dairy (18% of US outbreak vehicles), and untreated water (PMID: 42720049). Pork accounted for 46% of identified US outbreak vehicles.

Reservoirs. Swine are the primary reservoir (bioserotype 4/O:3); wildlife (badgers, roe deer, mustelids), poultry, and other livestock also carry the organism (PMID: 40582279, PMID: 41527869).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating exposure → clinical manifestation)

1. **Ingestion** of *Y. enterocolitica* in contaminated pork/dairy/water **leads to** delivery of viable bacteria to the small intestine (survival aided by psychrotolerance and acid tolerance).
2. Bacteria **colonize the terminal ileum**; expression of chromosomal adhesins/invasins is favored at intestinal temperature/pH.
3. **Invasin binds host β 1 integrins** on the apical surface of **M cells** overlying Peyer's patches, **resulting in** receptor-mediated internalization and translocation across the follicle-associated epithelium (PMID: 27107739, PMID: 26731748). (*Invasin is rapidly degraded by gut proteases, which limits the efficiency of this step — PMID: 21501502.*)
4. Uptake by **lamina propria CD103⁺ dendritic cells and monocyte-derived phagocytes** (invasin-dependent; YadA dispensable) **leads to** dissemination to mesenteric lymph nodes, and in some hosts the spleen and liver (PMID: 27107739).
5. At 37 °C, the **VirF/LcrF regulator induces the pYV-encoded T3SS**, which upon host-cell contact **injects Yop effectors** (YopH, YopE, YopT, YopO, YopP/J, YopM) into phagocytes (PMID: 8709853).
6. Yop effectors **subvert innate immunity**: YopH/E/T/O disrupt phagocytosis and actin dynamics; **YopP/YopJ inhibits NF- κ B/MAPK survival pathways, causing macrophage/DC apoptosis** and modulating caspase-1/IL-1 β (PMID: 22563435, PMID: 18559430). This **results**

in immune evasion and local tissue inflammation → **enterocolitis / terminal ileitis / mesenteric adenitis / pseudoappendicitis**.

7. **YstA heat-stable enterotoxin** activates guanylate cyclase in enterocytes, **contributing to** secretory diarrhea ([PMID: 28400007](#), [PMID: 30961808](#)).

Branch A — host iron status → septicemia. Because *Y. enterocolitica* lacks a high-affinity siderophore, **iron overload or deferoxamine therapy supplies chelated iron (ferrioxamine)**, which **leads to** unrestrained bacterial replication and **systemic septicemia with hepatic/splenic abscesses** ([PMID: 12019084](#), [PMID: 31219970](#)).

Branch B — post-infectious autoimmunity. In HLA-B27⁺ hosts, enteric infection **triggers reactive arthritis** ([PMID: 7221767](#)). Independently, **molecular mimicry** between the bacterial outer-membrane porin **OmpF (aa190–197)** and a leucine-rich domain of the **TSH receptor** generates cross-reactive, thyroid-stimulating antibodies, **contributing to Graves' disease / autoimmune thyroid disease** ([PMID: 20484489](#), [PMID: 32564766](#)).

Mechanistic categories

- **Molecular pathways:** T3SS/Yop-mediated inhibition of **NF-κB and MAPK** signaling; modulation of the **pyrin and caspase-1 inflammasomes** ([PMID: 31456795](#), [PMID: 22563435](#)); integrin/FAK signaling on invasion ([PMID: 26731748](#)). GO suggestions: GO:0030260 (entry into host cell), GO:0052031 (modulation by symbiont of host defense response), GO:0043123 (positive regulation of NF-κB signaling — inhibited).
- **Cellular processes:** macrophage/DC **apoptosis**, inhibited phagocytosis, inflammation, MDSC-mediated immunosuppression ([PMID: 39529636](#)).
- **Protein dysfunction (host):** Yop-driven dephosphorylation (YopH phosphatase) and cytoskeletal GTPase inactivation.
- **Immune involvement:** ILC3-mediated mucosal protection ([PMID: 32147792](#)); neutrophils resist YopJ/P-induced apoptosis ([PMID: 20174624](#)); autoimmunity via mimicry (§5).
- **Cell types (CL):** M cell (CL:0000682), dendritic cell (CL:0000451), macrophage (CL:0000235), neutrophil (CL:0000775), enterocyte (CL:0000584).
- **CHEBI entities:** iron (CHEBI:18248), deferoxamine (CHEBI:4356), ferrioxamine B (CHEBI:87109).

7. Anatomical Structures Affected

- **Primary organs/systems (digestive):** terminal **ileum** (UBERON:0002116), **Peyer's patches** (UBERON:0001211), **mesenteric lymph nodes** (UBERON:0002509), **appendix/cecum** region (pseudoappendicitis) ([PMID: 15254824](#)).
 - **Secondary organ involvement:** **liver** (UBERON:0002107) and **spleen** (UBERON:0002106) abscesses in septicemia; joints (reactive arthritis); skin (erythema nodosum); eye (uveitis); thyroid gland (Graves'); rarely heart (myopericarditis — [PMID: 35728146](#)) and inner ear/vestibular system ([PMID: 9376035](#)).
 - **Tissue/cell level:** intestinal **follicle-associated epithelium** and M cells; lymphoid tissue; mononuclear phagocytes.
 - **Subcellular (GO cellular component):** host plasma membrane / β 1-integrin complex (GO:0008305); host cytoskeleton (targeted by Yops); bacterial T3SS injectisome.
 - **Lateralization:** reactive arthritis is characteristically **asymmetric**, lower-extremity oligoarthritis.
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8. Temporal Development

- **Onset:** acute, typically 4–7 days after ingestion; enterocolitis predominates in infants/young children, pseudoappendicitis in older children/young adults ([PMID: 15254824](#)).
 - **Progression / duration:** enterocolitis is usually **self-limiting** over 1–3 weeks; diarrhea can occasionally persist. Septicemia is acute and can be fulminant, especially with rapid decompensation after transfusion in iron-overloaded patients ([PMID: 31219970](#)).
 - **Post-infectious window:** reactive arthritis and erythema nodosum appear **days to weeks** after the enteric phase; autoimmune thyroid sequelae are more delayed.
 - **Course pattern:** acute self-limited enteric phase; post-infectious syndromes may be episodic/relapsing. **Critical intervention windows:** early antibiotics in high-risk (iron-overloaded/immunocompromised) hosts; avoidance of deferoxamine during suspected infection.
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9. Inheritance and Population (Epidemiology)

- **Not heritable** — infectious etiology; no Mendelian inheritance, penetrance, founder effects, or carrier frequency apply. Host susceptibility to sequelae is modulated by HLA-B27.
 - **Incidence trend:** non-pestis yersiniosis had **the largest increase in reported incidence of any bacterial enteric infection in the US during 2016–2023** (PMID: 42720049).
 - **Reservoir prevalence:** *Y. enterocolitica* isolated from **9.31%** of pig tonsils (bioserotype 4/O:3 = 95.74%) (PMID: 29307129); ~9.3% porcine tonsils and 3.3% pig faeces in other surveys (PMID: 19058737); ~9.3–9.7% prevalence in retail meats (PMID: 41527869, PMID: 40382034).
 - **Geographic distribution:** worldwide in temperate/cold climates; bioserotype 4/O:3 predominant in Europe/Asia; highly pathogenic 1B/O:8 historically North American but now reported in Iran (PMID: 40669762).
 - **Age distribution:** enterocolitis skews to young children (<5 years — PMID: 40669762); pseudoappendicitis to older children/young adults.
 - **Seasonality:** cold-season predominance consistent with psychrotolerance (PMID: 40669762).
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10. Diagnostics

Culture (gold standard). Stool culture on **CIN (cefsulodin–irgasan–novobiocin) agar** with **cold enrichment** (4 °C) enhances recovery — cold enrichment recovered ~25% of 4/O:3 and 2/O:9 strains not otherwise detected (PMID: 19219471). Blood culture in suspected septicemia.

Bio/serotyping. Congo-red magnesium-oxalate (**CR-MOX**) agar detects the pYV virulence plasmid; biochemical panels assign biotype; agglutination/PCR serotyping identifies O:3, O:9, O:8, O:5,27 (PMID: 19219471, PMID: 40445798).

Molecular confirmation. PCR for virulence genes (**ail**, **ystA/ystB**, **virF**, **yadA**, **inv**, **myfA**) distinguishes pathogenic from non-pathogenic (biotype 1A) strains; 16S rRNA and *gyrB* sequencing resolves *Y. enterocolitica*-like species; multiplex/quadruplex RT-qPCR patho-serotyping is available (PMID: 19219471, PMID: 40445798).

Serology. Anti-*Yersinia* antibody titers (e.g., against O:3, O:9) support diagnosis of post-infectious syndromes such as reactive arthritis (PMID: 9376035). Caveat: O:9 cross-reacts serologically with *Brucella*.

Imaging. CT/ultrasound in pseudoappendicitis shows terminal ileitis and mesenteric lymphadenopathy (helping avoid unnecessary appendectomy).

Differential diagnosis. Acute appendicitis, Crohn's disease/terminal ileitis, *Campylobacter*/*Salmonella*/*Shigella* enterocolitis, mesenteric adenitis of other cause.

11. Outcome / Prognosis

- **Enterocolitis:** excellent prognosis; usually **self-limiting** without antibiotics.
- **Pseudoappendicitis:** favorable; risk of unnecessary surgery.
- **Septicemia:** serious, with mortality concentrated in iron-overloaded/immunocompromised patients; requires prompt IV antibiotics (PMID: 21266629, PMID: 31219970). Case reports document successful treatment (e.g., 14-day ceftriaxone) with resolution and no recurrence.
- **Post-infectious morbidity:** reactive arthritis (ReA overall prevalence ~1/1000) can cause weeks-to-months of joint disability (PMID: 40473988); persistent extraintestinal manifestations (uveitis, erythema nodosum, rare vestibular loss/myopericarditis) (PMID: 9376035, PMID: 35728146).
- **Prognostic factors:** host iron status, immunocompetence, age, timeliness of antibiotic therapy, and infecting bioserotype (1B/O:8 highly pathogenic).

12. Treatment

General principle. Uncomplicated enterocolitis is **self-limiting** and generally does **not** require antibiotics; supportive care (rehydration) suffices. Antibiotics are indicated for **invasive/septicemic disease** and in high-risk hosts.

Antimicrobial susceptibility profile.

Antimicrobial class	Susceptibility	Evidence
Ampicillin, 1st-gen cephalosporins (cefazolin)	Intrinsically RESISTANT (chromosomal β -lactamases blaA/blaB)	PMID: 10803262 , PMID: 10755244
3rd-gen cephalosporins (cefotaxime, ceftriaxone)	Susceptible — drug of choice	PMID: 10803262 , PMID: 21266629
Fluoroquinolones (ciprofloxacin, ofloxacin)	Susceptible; highly effective in models	PMID: 10803262 , PMID: 1952849
Aminoglycosides (gentamicin)	Susceptible	PMID: 10755244 , PMID: 20828475
TMP-SMX, tetracyclines, chloramphenicol, imipenem, aztreonam	Susceptible	PMID: 10755244

Susceptibility is independent of the pYV plasmid ([PMID: 10755244](#)). In murine systemic infection, newer β -lactams may fail while fluoroquinolones (ofloxacin, 5 mg/kg) are very effective ([PMID: 1952849](#)).

Regimens. Septicemia is typically treated with a **third-generation cephalosporin** (e.g., ceftriaxone) $\pm$ an **aminoglycoside** or a **fluoroquinolone**; combination therapy is used for severe disease. NCIT term suggestions: Ceftriaxone (NCIT:C749), Ciprofloxacin (NCIT:C405), Gentamicin (NCIT:C557), Trimethoprim-Sulfamethoxazole (NCIT:C273).

Critical host management. In iron-overloaded patients, **discontinue deferoxamine** during infection and consider deferiprone/deferiasirox, which do not promote bacterial growth ([PMID: 12019084](#), [PMID: 19413741](#)).

AMR surveillance. Prudent antimicrobial use in pig farming is emphasized to limit resistance-gene spread via horizontal transfer ([PMID: 41413462](#), [PMID: 35369517](#)).

Advanced/experimental therapeutics. Not applicable — no gene/cell/RNA/targeted therapies; treatment is antimicrobial and supportive.

13. Prevention

- **Primary prevention (food-chain hygiene — the mainstay):** "Good hygienic slaughter practices are essential to prevent contamination of pork" (PMID: 19058737). Thorough cooking of pork, pasteurization of dairy, avoidance of cross-contamination, on-farm biosecurity, and cold-chain management from farm to fork (PMID: 40023561). Key on-farm risk factors to mitigate: batch mixing in fattening herds and poor pen cleaning/biosecurity (PMID: 19175574).
 - **Immunization: No licensed human vaccine exists.** Oral mucosal-vaccine constructs using *Y. enterocolitica* OmpH as a carrier are experimental (PMID: 32569608).
 - **Secondary prevention:** microbiological surveillance of the pork chain; molecular subtyping (SNP/MLST) to track transmission hotspots (slaughterhouses, markets) (PMID: 41539755).
 - **Tertiary prevention:** in iron-overloaded/dialysis patients, monitor ferritin, avoid deferoxamine when infection is suspected, and screen blood products (transfusion transmission hazard) (PMID: 21266629, PMID: 33341965).
 - **Public health:** food-safety education, blood-bank visual inspection/interdiction of contaminated RBC units.
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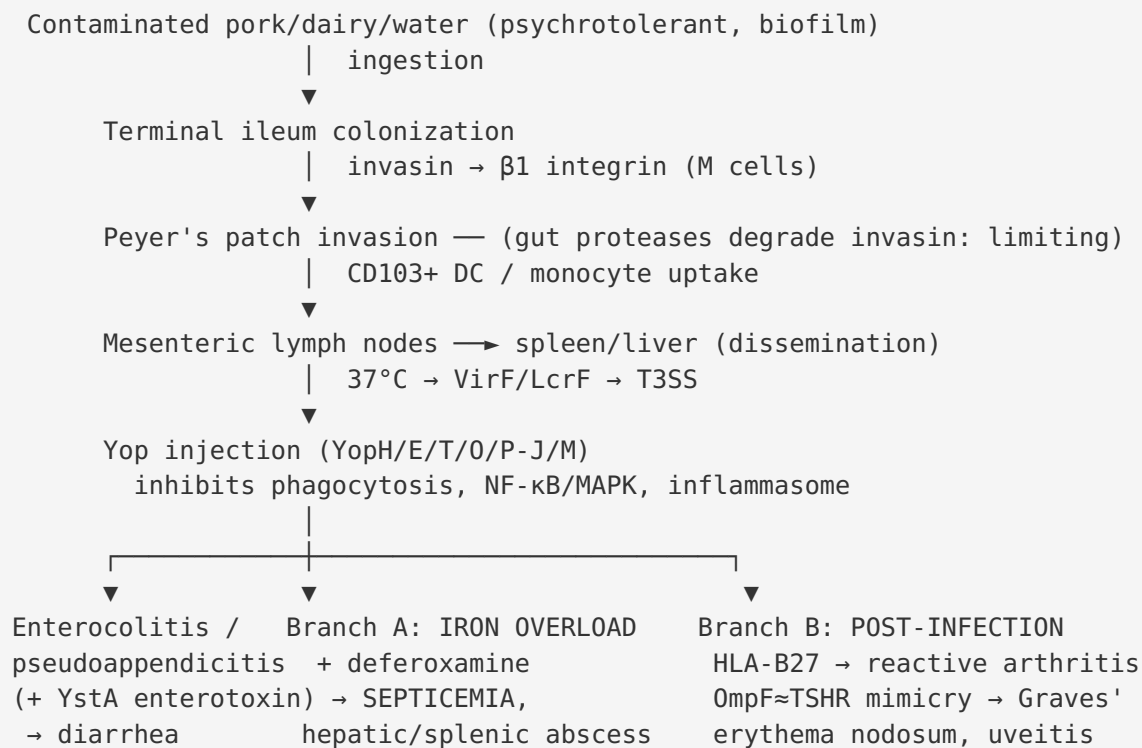
14. Other Species / Natural Disease

- **Taxonomy of affected/reservoir species:** *Sus scrofa domesticus* (pig; NCBI:txid9825) — primary reservoir; also cattle, sheep, poultry (chicken, duck), dogs, cats, and wildlife (badgers, roe deer, mustelids) (PMID: 40582279, PMID: 41527869).
- **Zoonotic transmission:** fecal–oral via contaminated food, water, and animal contact; strong swine–human genomic link established by pangenome analysis (PMID: 41413462). Environmental persistence broadens the ecology.
- **Comparative pathology:** pigs are typically **asymptomatic carriers** (tonsils, gut, mesenteric nodes), unlike humans who develop enteric disease — an important distinction for control.
- **Evolutionary conservation:** T3SS/Yop virulence mechanisms are conserved across pathogenic *Yersinia* (*Y. enterocolitica*, *Y. pseudotuberculosis*, *Y. pestis*), enabling cross-species mechanistic inference (PMID: 18559430, PMID: 38271464).

15. Model Organisms

- **Rodent oral-infection models faithfully recapitulate human yersiniosis:** "experimental *Y. enterocolitica* infection in rodents resembles yersiniosis in humans and thus offers extraordinary opportunities to study the sequential steps of the infectious process" (PMID: 8363822).
 - **Mouse (C57BL/6):** oral infection with serotype O:8 strains (WA-314, WAP-314) models Peyer's-patch/M-cell entry, ILC3-mediated protection (PMID: 32147792), NO-producing Mo-MDSC immunosuppression (PMID: 39529636), and bacterial population dynamics (PMID: 35205164). The deferoxamine-potential LD₅₀ experiment was performed in mice (PMID: 12019084).
 - **Rabbit:** models enterotoxin-mediated enteritis and humoral response (PMID: 8363822).
 - **In vitro / cellular:** Caco-2 intestinal epithelial cells for invasin- β 1-integrin internalization (PMID: 26731748); primary macrophages/neutrophils for Yop-mediated apoptosis studies (PMID: 22563435, PMID: 20174624); suckling-mouse bioassay for YstA enterotoxin activity (PMID: 39805396).
 - **Computational:** ODE-based models of gastrointestinal bacterial population dynamics (PMID: 35205164).
 - **Limitations:** mouse strains used experimentally are often serotype O:8 (highly pathogenic 1B), which may not reflect the globally dominant 4/O:3; post-infectious autoimmune sequelae (Graves', HLA-B27 ReA) are not well captured in rodents.
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Mechanistic Model / Integrated Interpretation



Upstream vs downstream. Invasin-mediated M-cell entry is the upstream initiating event; T3SS/Yop immune subversion is the central amplifying node; enterocolitis, septicemia, and autoimmune sequelae are downstream outcomes gated by host iron status and HLA background.

Evidence Base (key literature)

PMID	Contribution	Type
26403101	Swine/pork as reservoir & main infection source	Review
29307129	9.31% pig-tonsil prevalence; 4/0:3 = 95.74%	Surveillance
12019084	Deferoxamine lowers LD ₅₀ >5 log; deferiprone does not	Model organism
31219970	Transfusional iron overload → fulminant yersiniosis	Case report
27107739	Invasin-dependent DC/M-cell uptake & dissemination	Mechanistic
22563435	YopJ inhibits survival pathways → macrophage death	Mechanistic
40473988	<i>Y. enterocolitica</i> as post-enteric ReA trigger	Review
7221767	ReA linked to HLA-B27	Case report
32564766	Graves' association OR 6.12 (3.71–10.10)	Meta-analysis
20484489	OmpF–TSHR molecular mimicry	In vitro
42720049	Rising US incidence; pork 46%, dairy 18%	Surveillance
19219471	Culture/cold-enrichment diagnostics; bioserotype prevalence	Surveillance
10803262	3rd-gen cephalosporins & ciprofloxacin most active	In vitro
1952849	Fluoroquinolone efficacy vs β-lactam failure	Model organism
15254824	Three clinical syndromes defined	Review
19058737	Slaughter hygiene as key prevention	Surveillance
8363822	Rodent model fidelity to human disease	Review
28400007	ystA best pathogenicity marker; ystB for 1A	Molecular
28283072	ail as main chromosomal virulence marker	Molecular
41539755	Psychrotolerance/biofilm across cold chain	Surveillance

Limitations and Knowledge Gaps

- **No human genetic causal architecture** — as an infectious disease, sections on causal genes, pathogenic ACMG variants, inheritance, penetrance, and carrier frequency are **not applicable**; only host modifiers (HLA-B27, iron-handling) are relevant.
- **Quantitative human epidemiology is incomplete** here — precise population incidence/prevalence per 100,000 and absolute mortality rates were not extracted; the strongest quantitative signal is the documented US incidence *increase* rather than an absolute rate.
- **Causal certainty of autoimmune sequelae** — the Graves'/AITD association is epidemiological (OR-based meta-analysis) plus a plausible mimicry mechanism; direct causation in humans remains inferential.
- **Model–strain mismatch** — much mechanistic work uses highly pathogenic 1B/O:8 mouse strains, whereas most human disease is 4/O:3; findings may not fully transfer.
- **Effect sizes for some clinical claims** (e.g., proportion progressing to septicemia, ReA incidence specifically after *Yersinia*) rely on case series and reviews rather than large cohorts.

Proposed Follow-up Actions

1. **Extract absolute incidence/prevalence** from GBD, CDC FoodNet, and Orphanet to complete Section 9 quantitatively.
 2. **Systematic review of HLA-B27 penetrance** for *Yersinia*-triggered reactive arthritis (attributable risk, time-to-onset).
 3. **Prospective study** of oral iron-chelator choice (deferiprone/deferiasirox vs deferoxamine) on yersiniosis incidence in transfusion-dependent patients.
 4. **Genomic surveillance (One Health)** linking swine and human isolates via SNP/MLST to quantify attributable transmission and AMR gene flow ([PMID: 41413462](#)).
 5. **4/O:3-based infection models** to align mechanistic studies with the globally dominant human bioserotype.
 6. **Mechanistic test of OmpF–TSHR mimicry in vivo** to establish causality for the Graves' association.
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Report compiled from 13 confirmed findings across 73 reviewed papers over 5 investigation iterations. Evidence types are labeled (human clinical, model organism, in vitro, computational, surveillance/review) throughout.

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